WORKED SOLUTION

STEP III 1995, Question 3

Pure • Damped Harmonic Motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q3

The key idea

Try $x = \mathrm{e}^{\lambda t}$: the differential equation becomes a quadratic in λ, and the three cases are just the three possibilities for its discriminant (two real roots, a repeated root, complex roots). For the damped case, the stationary points come round at equal time intervals of half a period, and over each such interval the oscillating factor simply flips sign while the exponential factor shrinks by the same amount. That is why the ratio of successive extremes is constant.

Solution

Substituting $x = \mathrm{e}^{\lambda t}$ gives the auxiliary equation

$$\lambda^2 + 2k\lambda + 1 = 0, \qquad \lambda = -k \pm \sqrt{k^2 - 1}.$$

Throughout, A and B are arbitrary constants.

Part (i): $k > 1$

The roots are real and distinct, so

$$x = A\,\mathrm{e}^{(-k+\sqrt{k^2-1})\,t} + B\,\mathrm{e}^{(-k-\sqrt{k^2-1})\,t}.$$

(Both exponents are negative, since $\sqrt{k^2-1} < k$: this is *overdamping*, with no oscillation.)

Part (ii): $k = 1$

The auxiliary equation is $(\lambda+1)^2 = 0$, a repeated root $\lambda = -1$. The second independent solution is $t\,\mathrm{e}^{-t}$, so

$$x = (A + Bt)\,\mathrm{e}^{-t}.$$

(Check: if $x = te^{-t}$ then $x' = (1-t)e^{-t}$ and $x'' = (t-2)e^{-t}$, and $(t-2) + 2(1-t) + t = 0$.)

Part (iii): $0 < k < 1$

Now $k^2 - 1 < 0$, so the roots are $\lambda = -k \pm \mathrm{i}\omega$ where

$$\omega = \sqrt{1 - k^2} > 0.$$

Taking real combinations of $\mathrm{e}^{(-k \pm \mathrm{i}\omega)t}$,

$$x = \mathrm{e}^{-kt}(A\cos\omega t + B\sin\omega t).$$

The successive extremes

With $x(0) = 0$ we need $A = 0$, so $x = B\,\mathrm{e}^{-kt}\sin\omega t$, and $B \neq 0$ (otherwise $x \equiv 0$ and there are no maxima or minima). Differentiating,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = B\,\mathrm{e}^{-kt}(\omega\cos\omega t - k\sin\omega t).$$

Since $k^2 + \omega^2 = 1$, we can write $k = \sin\varphi$, $\omega = \cos\varphi$ with $0 < \varphi < \frac{\pi}{2}$, and then

$$\omega\cos\omega t - k\sin\omega t = \cos(\omega t + \varphi).$$

So $\dfrac{\mathrm{d}x}{\mathrm{d}t} = B\,\mathrm{e}^{-kt}\cos(\omega t + \varphi)$, which is zero exactly when $\omega t + \varphi$ is an odd multiple of $\frac{\pi}{2}$, and changes sign at each such time. The stationary points (for $t > 0$) are therefore at

$$t_n = t_1 + \frac{(n-1)\pi}{\omega}, \qquad n = 1, 2, 3, \ldots,$$

equally spaced π/ω apart, and since $\mathrm{d}x/\mathrm{d}t$ changes sign at each one they really are alternately maxima and minima.

Ask yourself How do we compare $x(t_{n+1})$ with $x(t_n)$ without finding t_1 explicitly?

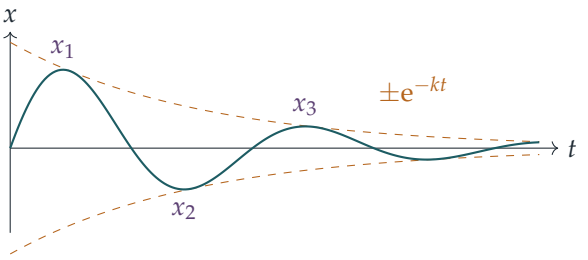
Moving on by π/ω in time adds π to ωt, and $\sin(\theta + \pi) = -\sin\theta$. So

$$x_{n+1} = B\,\mathrm{e}^{-k(t_n + \pi/\omega)}\sin(\omega t_n + \pi) = -\,\mathrm{e}^{-k\pi/\omega}\,B\,\mathrm{e}^{-kt_n}\sin\omega t_n = -\,\mathrm{e}^{-k\pi/\omega}\,x_n.$$

Also $x_n \neq 0$, because $\sin\omega t_n = 0$ would force $\cos\omega t_n = \pm 1$ and then $\mathrm{d}x/\mathrm{d}t = \pm B\omega\,\mathrm{e}^{-kt_n} \neq 0$. Hence

$$\left|\frac{x_{n+1}}{x_n}\right| = \alpha, \qquad \text{where } \alpha = \mathrm{e}^{-k\pi/\omega} = \exp\left(-\frac{k\pi}{\sqrt{1-k^2}}\right),$$

which does not depend on n, as required. (The minus sign confirms the alternation between maxima and minima.)



The formula for k^2

Taking logarithms, $\ln\alpha = -\dfrac{k\pi}{\sqrt{1-k^2}}$, so

$$(\ln\alpha)^2 = \frac{k^2\pi^2}{1-k^2} \implies (\ln\alpha)^2 - k^2(\ln\alpha)^2 = k^2\pi^2 \implies k^2 = \frac{(\ln\alpha)^2}{\pi^2 + (\ln\alpha)^2},$$

which is the required result.

Checks

- As $k \to 0$ (no damping), $\alpha \to 1$ and $\ln\alpha \to 0$, so the formula gives $k^2 \to 0$, as it should: the extremes all have the same size.
- As $k \to 1^-$, $\alpha \to 0$ and $(\ln\alpha)^2 \to \infty$, so $k^2 \to 1$: at critical damping the motion no longer swings back.
- Numerically, with $k = 0.5$ and $x(0) = 0$, stepping the equation forward gives successive extreme ratios 0.163034, matching $\mathrm{e}^{-0.5\pi/\sqrt{0.75}} \approx 0.163034$; and then $(\ln\alpha)^2/(\pi^2 + (\ln\alpha)^2) = 0.25 = k^2$.

Common slips

- Taking the time between a maximum and the next minimum to be a full period $2\pi/\omega$ rather than half a period π/ω; this gives α^2 instead of α.
- Thinking the extremes occur where $\sin\omega t = \pm 1$. They don't: the decaying factor shifts them earlier, to where $\tan\omega t = \omega/k$. The argument above never needs their exact position, only their spacing.

Beyond STEP

The quantity $-\ln\alpha$ is the *logarithmic decrement*: engineers measure it from a recorded oscillation, and this question's formula turns it into the damping k. Boyce and DiPrima, *Elementary Differential Equations*, treats over-, critical and under-damping in detail.

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